Unit D Lesson 1 Review

Multiple Choice Bank — 75 Questions

Question Set

MC 1. According to the Hardy-Weinberg principle, what does the equation $p^2 + 2pq + q^2 = 1$ represent?

A. The sum of allele frequencies in a population
B. The sum of genotype frequencies in a population
C. The rate of mutation in a population
D. The change in allele frequencies over time

Correct answer: B

Rationale: The equation $p^2 + 2pq + q^2 = 1$ represents the sum of all genotype frequencies in a population, where p^2 is the frequency of homozygous dominant individuals, $2pq$ is the frequency of heterozygotes, and q^2 is the frequency of homozygous recessive individuals.

MC 2. In a population of 1000 individuals where 360 show a recessive phenotype, what is the frequency of the dominant allele (p)?

A. 0.36
B. 0.60
C. 0.40
D. 0.64

Correct answer: C

Rationale: With 360 recessive individuals out of 1000, $q^2 = 0.36$, so $q = 0.6$. Since $p + q = 1$, $p = 1 - 0.6 = 0.4$.

MC 3. Which condition is NOT required for Hardy-Weinberg equilibrium?

A. Large population size
B. Random mating
C. Equal allele frequencies
D. No mutations

Correct answer: C

Rationale: Hardy-Weinberg equilibrium does not require equal allele frequencies. The five conditions are: no mutations, no gene flow, large population size, random mating, and no natural selection.

MC 4. What does the Hardy-Weinberg principle function as in population genetics?

A. A proof of evolution
B. A mathematical null hypothesis
C. A method to create mutations
D. A way to eliminate genetic variation

Correct answer: B

Rationale: The Hardy-Weinberg principle functions as a mathematical null hypothesis for population genetics, establishing what happens to allele frequencies when no evolutionary forces are acting.

MC 5. In the equation $p + q = 1$, what do p and q represent?

A. Genotype frequencies
B. Population sizes
C. Allele frequencies in a two-allele system
D. Mutation rates

Correct answer: C

Rationale: In a two-allele system, p and q represent the frequencies of the two alleles, which must sum to 1 (representing 100% of all alleles for that gene).

MC 6. What type of growth occurs when resources are unlimited according to the lesson material?

A. Linear growth
B. Logistic growth
C. Exponential growth
D. Zero growth

Correct answer: C

Rationale: The material states that populations grow exponentially when resources are unlimited, following the equation $dN/dt = rN$.

MC 7. What does the term '2pq' represent in Hardy-Weinberg calculations?

A. Frequency of homozygous dominant individuals
B. Frequency of heterozygous individuals
C. Frequency of homozygous recessive individuals
D. Total population size

Correct answer: B

Rationale: In the Hardy-Weinberg equation, 2pq represents the frequency of heterozygous individuals in the population.

MC 8. What happens to genotype frequencies during random mating according to Hardy-Weinberg?

A. They change randomly each generation
B. They remain constant across generations
C. They always increase
D. They always decrease

Correct answer: B

Rationale: Under Hardy-Weinberg equilibrium conditions with random mating, both allele frequencies and genotype frequencies remain constant across generations.

MC 9. Which factor would cause deviation from Hardy-Weinberg equilibrium?

A. Large population size
B. Random mating
C. Gene flow between populations
D. Absence of mutations

Correct answer: C

Rationale: Gene flow between populations introduces new alleles or changes allele frequencies, violating one of the five conditions required for Hardy-Weinberg equilibrium.

MC 10. In logistic growth, what does 'K' represent?

A. Growth rate
B. Carrying capacity
C. Birth rate
D. Death rate

Correct answer: B

Rationale: In the logistic growth equation $dN/dt = rN(K-N)/K$, K represents the carrying capacity—the maximum sustainable population size.

MC 11. What creates the probability of forming an AA genotype during random mating?

A. $p \times q$

B. $q \times q$
C. $p \times p$
D. $2 \times p \times q$

Correct answer: C

Rationale: The probability of forming an AA genotype equals $p \times p = p^2$, where p is the frequency of allele A in the gene pool.

MC 12. According to the material, what type of competition occurs between members of the same species?

A. Interspecific competition
B. Intraspecific competition
C. Exploitation competition
D. Apparent competition

Correct answer: B

Rationale: The material describes intraspecific competition as occurring between members of the same species for limited resources.

MC 13. What prevents genetic drift from affecting allele frequencies?

A. Small population size
B. Nonrandom mating
C. Large population size
D. Natural selection

Correct answer: C

Rationale: Large population size prevents genetic drift (random sampling effects) from changing allele frequencies, which is one of the five conditions for Hardy-Weinberg equilibrium.

MC 14. In a Drosophila population with $p = q = 0.5$ for eye color alleles, what percentage of flies should be heterozygous according to Hardy-Weinberg?

A. 25%
B. 50%
C. 75%
D. 100%

Correct answer: B

Rationale: With $p = q = 0.5$, the frequency of heterozygotes = $2pq = 2(0.5)(0.5) = 0.5$ or 50%.

MC 15. What does Hardy-Weinberg equilibrium represent in terms of evolution?

A. Rapid evolutionary change
B. No evolutionary forces acting on the population
C. Strong natural selection
D. High mutation rates

Correct answer: B

Rationale: Hardy-Weinberg equilibrium represents a state where no evolutionary forces are acting on a population, serving as a baseline for detecting evolution.

MC 16. According to the material, what type of growth pattern shows an S-shaped curve?

A. Exponential growth
B. Linear growth
C. Logistic growth
D. Irregular growth

Correct answer: C

Rationale: Logistic growth creates an S-shaped curve as population growth slows when approaching carrying capacity.

MC 17. What happens to allele frequencies when natural selection favors particular alleles?

A. They remain constant
B. They change by affecting reproductive success
C. They become equal
D. They disappear completely

Correct answer: B

Rationale: Natural selection changes allele frequencies by affecting reproductive success of individuals carrying different alleles.

MC 18. In the predator-prey relationship described, what happens when prey density is low?

A. Predator populations increase rapidly
B. Predator populations crash due to lack of food
C. Prey populations explode immediately
D. Both populations remain stable

Correct answer: B

Rationale: The material states that when prey density is low, predator populations crash due to lack of food.

MC 19. What mechanism explains Hardy-Weinberg stability during sexual reproduction?

A. Selective mating preferences
B. Random combination of alleles during gamete formation and fertilization
C. Directed mutations
D. Environmental pressures

Correct answer: B

Rationale: Hardy-Weinberg stability occurs through random combination of alleles when gametes form through meiosis and combine during fertilization.

MC 20. What type of reproductive strategy involves producing many offspring with minimal parental care?

A. K-selected strategy
B. r-selected strategy
C. Equilibrium strategy
D. Climax strategy

Correct answer: B

Rationale: r-selected species produce many offspring with minimal parental investment, maximizing reproductive rate.

MC 21. If $q^2 = 0.36$ in a population, what is the value of q?

A. 0.36
B. 0.40
C. 0.60
D. 0.64

Correct answer: C

Rationale: If $q^2 = 0.36$, then $q = \sqrt{0.36} = 0.6$.

MC 22. According to the material, what drives primary succession?

A. Human interference
B. Pioneer species colonizing bare substrate

C. Large herbivores
D. Predator removal

Correct answer: B

Rationale: Primary succession is driven by pioneer species colonizing bare substrate like volcanic rock or glacial till.

MC 23. What effect does nonrandom mating have on a population?

A. It changes allele frequencies immediately
B. It alters genotype frequencies even if allele frequencies remain stable
C. It increases mutation rates
D. It eliminates all genetic variation

Correct answer: B

Rationale: Nonrandom mating alters genotype frequencies even if allele frequencies remain stable, violating Hardy-Weinberg equilibrium.

MC 24. In the competitive exclusion principle, what happens when two species have identical niches?

A. Both species thrive equally
B. They merge into one species
C. One species eliminates the other
D. They share resources equally

Correct answer: C

Rationale: The competitive exclusion principle states that when two species have identical niches, one will eliminate the other through superior competitive ability.

MC 25. What does the term 'genetic equilibrium' mean in Hardy-Weinberg context?

A. All alleles have equal frequency
B. Allele and genotype frequencies remain constant across generations
C. No genetic variation exists
D. Evolution is occurring rapidly

Correct answer: B

Rationale: Genetic equilibrium is a state where both allele frequencies and genotype frequencies remain constant across generations.

MC 26. According to the material, what characterizes mutualistic relationships?

A. One species benefits, one is harmed
B. Both species are harmed
C. Both species benefit
D. One species benefits, one is unaffected

Correct answer: C

Rationale: Mutualistic relationships are characterized by both species benefiting from the interaction.

MC 27. What happens during the lag phase of population growth?

A. Population declines rapidly
B. Population grows at maximum rate
C. Population grows slowly as individuals establish
D. Population reaches carrying capacity

Correct answer: C

Rationale: During the lag phase, population grows slowly as individuals establish and begin reproducing.

MC 28. How do mutations affect Hardy-Weinberg equilibrium?

A. They maintain equilibrium
B. They introduce new alleles or alter existing ones
C. They have no effect
D. They only affect phenotypes

Correct answer: B

Rationale: Mutations introduce new alleles or alter existing ones, changing allele frequencies and violating Hardy-Weinberg equilibrium.

MC 29. What type of species typically shows Type III survivorship curves?

A. Large mammals with extensive parental care
B. Species with constant mortality throughout life
C. Species producing many offspring with high early mortality
D. Species with low birth rates

Correct answer: C

Rationale: Type III survivorship curves are shown by species that produce many offspring with high early mortality, like many fish and plants.

MC 30. In the equation $dN/dt = rN$, what does 'r' represent?

A. Carrying capacity
B. Per capita growth rate
C. Population size
D. Death rate only

Correct answer: B

Rationale: In the exponential growth equation, 'r' represents the per capita growth rate (births minus deaths per individual).

MC 31. What creates the $(K-N)/K$ term in logistic growth?

A. Unlimited resources
B. Environmental resistance as population approaches carrying capacity
C. Increased birth rate
D. Decreased death rate

Correct answer: B

Rationale: The $(K-N)/K$ term represents environmental resistance that increases as population size (N) approaches carrying capacity (K).

MC 32. According to the material, what occurs during secondary succession?

A. Colonization of bare rock
B. Recovery after disturbance with soil present
C. Formation of new islands
D. Glacier formation

Correct answer: B

Rationale: Secondary succession occurs after disturbance when soil remains, allowing faster recovery than primary succession.

MC 33. What does each gamete receive during meiosis according to Hardy-Weinberg mechanics?

A. Two allele copies from each parent
B. One allele copy from each parent
C. All alleles from one parent

D. Random numbers of alleles

Correct answer: B

Rationale: During meiosis, each gamete receives one allele copy from each parent, which is fundamental to Hardy-Weinberg mechanics.

MC 34. What characterizes K-selected species according to the material?

A. Many offspring with minimal care
B. Few offspring with substantial parental investment
C. No parental care
D. Extremely high growth rates

Correct answer: B

Rationale: K-selected species produce few offspring with substantial parental investment, adapted for stable environments near carrying capacity.

MC 35. What type of feedback occurs in predator-prey cycles?

A. Positive feedback only
B. No feedback
C. Negative feedback creating oscillations
D. Random feedback

Correct answer: C

Rationale: Predator-prey relationships involve negative feedback loops that create characteristic population oscillations.

MC 36. If $p = 0.4$ in a population, what is the frequency of heterozygotes?

A. 0.16
B. 0.36
C. 0.48
D. 0.60

Correct answer: C

Rationale: If $p = 0.4$, then $q = 0.6$. Frequency of heterozygotes = $2pq = 2(0.4)(0.6) = 0.48$.

MC 37. What allows researchers to test Hardy-Weinberg predictions in laboratory populations?

A. Inability to control conditions
B. Establishing populations with known initial allele frequencies
C. Random environmental changes
D. Uncontrolled mating

Correct answer: B

Rationale: Researchers can test Hardy-Weinberg predictions by establishing populations with known initial allele frequencies and tracking changes over generations.

MC 38. According to the material, what happens when a keystone species is removed?

A. No effect on community
B. Minor changes only
C. Disproportionate effects on community structure
D. Increased stability

Correct answer: C

Rationale: Keystone species have disproportionate effects on community structure relative to their abundance.

MC 39. What biological significance does the Hardy-Weinberg principle provide?

A. Proof that evolution never occurs
B. A baseline for detecting evolutionary forces
C. A way to prevent genetic change
D. Evidence that all populations evolve

Correct answer: B

Rationale: Hardy-Weinberg provides a mathematical baseline for detecting when evolutionary forces are acting on a population.

MC 40. In the marine protected area example, what is predicted to happen to genetic diversity?

A. Decrease due to isolation
B. Remain unchanged
C. Increase due to reduced fishing pressure
D. Disappear completely

Correct answer: C

Rationale: Protection from fishing pressure is predicted to increase genetic diversity in fish populations.

MC 41. What determines the probability of any specific genotype appearing in offspring under random mating?

A. Environmental conditions
B. Frequencies of contributing alleles in the gene pool
C. Parental phenotypes only
D. Mutation rates

Correct answer: B

Rationale: Under random mating, the probability of any specific genotype depends solely on the frequencies of the contributing alleles in the gene pool.

MC 42. According to the material, what creates demographic and genetic pressures?

A. Absence of interactions
B. Species interaction patterns
C. Stable environments only
D. Lack of competition

Correct answer: B

Rationale: Species interaction patterns (predator-prey, competitive, mutualistic) create specific demographic and genetic pressures.

MC 43. What type of survivorship curve do humans typically show?

A. Type I
B. Type II
C. Type III
D. Type IV

Correct answer: A

Rationale: Humans show Type I survivorship curves with low early mortality and increasing mortality in old age.

MC 44. What happens to population growth rate as N approaches K in logistic growth?

A. It increases
B. It remains constant
C. It approaches zero
D. It becomes negative

Correct answer: C

Rationale: As population size (N) approaches carrying capacity (K), the growth rate approaches zero due to environmental resistance.

MC 45. According to Hardy-Weinberg, what is required for allele frequencies to remain stable?

A. Only large population size
B. All five equilibrium conditions must be met
C. Just random mating
D. Only absence of mutations

Correct answer: B

Rationale: All five conditions (no mutations, no gene flow, large population, random mating, no selection) must be met for Hardy-Weinberg equilibrium.

MC 46. What characterizes the exponential growth phase?

A. Slow, steady increase
B. Population doubles at regular intervals
C. Immediate plateau
D. Declining growth rate

Correct answer: B

Rationale: During exponential growth, population doubles at regular intervals when resources are unlimited.

MC 47. In a two-allele system, why must $p + q = 1$?

A. It's an arbitrary rule
B. The frequencies of both alleles must sum to 100%
C. Populations always have equal allele frequencies
D. Mutations require this relationship

Correct answer: B

Rationale: In a two-allele system, $p + q = 1$ because the frequencies of both alleles must sum to 1, representing 100% of all alleles for that gene.

MC 48. What type of competition occurs between different species according to the material?

A. Intraspecific competition
B. Interspecific competition
C. Genetic competition
D. Temporal competition

Correct answer: B

Rationale: Interspecific competition occurs between different species for shared resources.

MC 49. What creates cascading effects across biological organization levels?

A. Isolation of each level
B. Changes at one level affecting other levels
C. Absence of environmental factors
D. Static populations

Correct answer: B

Rationale: Changes at one level create cascading effects at other levels through predictable mechanisms.

MC 50. According to the material, what represents coordinated trait combinations?

A. Random mutations
B. Life history strategies
C. Genetic drift

D. Gene flow

Correct answer: B

Rationale: Life history strategies represent coordinated trait combinations that optimize reproductive success under specific environmental conditions.

MC 51. What emerges automatically through random mating according to Hardy-Weinberg?

A. New alleles

B. Changed allele frequencies

C. Mathematical genotype frequency relationships

D. Natural selection

Correct answer: C

Rationale: The mathematical relationship between allele and genotype frequencies emerges automatically through random mating, requiring no special mechanisms.

MC 52. What phase follows the lag phase in population growth?

A. Death phase

B. Exponential growth phase

C. Immediate plateau

D. Decline phase

Correct answer: B

Rationale: The exponential growth phase follows the lag phase as reproduction accelerates.

MC 53. How does gene flow affect Hardy-Weinberg equilibrium?

A. It maintains equilibrium

B. It brings in alleles from other populations

C. It has no effect

D. It only affects phenotypes

Correct answer: B

Rationale: Gene flow brings in alleles from other populations, changing allele frequencies and violating equilibrium.

MC 54. What creates dynamic equilibria in ecological systems according to the material?

A. Static conditions

B. Feedback between environmental pressures and evolutionary change

C. Absence of species interactions

D. Constant population sizes

Correct answer: B

Rationale: Feedback between environmental pressures and evolutionary/demographic changes creates dynamic equilibria in ecological systems.

MC 55. In Hardy-Weinberg calculations, what does p^2 represent?

A. Frequency of heterozygotes

B. Frequency of homozygous dominant individuals

C. Frequency of homozygous recessive individuals

D. Total population size

Correct answer: B

Rationale: p^2 represents the frequency of homozygous dominant individuals in the population.

MC 56. According to the material, what affects survival probabilities?

A. Only genetic factors
B. Defense mechanisms through molecular and behavioral adaptations
C. Random chance alone
D. Population size only

Correct answer: B

Rationale: Defense mechanisms function through molecular and behavioral adaptations that affect survival probabilities.

MC 57. What allows small population sizes to change allele frequencies?

A. Natural selection
B. Random sampling effects (genetic drift)
C. Mutations
D. Gene flow

Correct answer: B

Rationale: Small population sizes allow random sampling effects (genetic drift) to change allele frequencies.

MC 58. What characterizes climax communities according to the material?

A. Rapid species turnover
B. Stable species composition
C. Only pioneer species
D. Absence of competition

Correct answer: B

Rationale: Climax communities are characterized by relatively stable species composition.

MC 59. During fertilization, what equals the probability of forming an aa genotype?

A. $p \times p$
B. $p \times q$
C. $q \times q$
D. $2pq$

Correct answer: C

Rationale: The probability of forming an aa genotype equals $q \times q = q^2$, where q is the frequency of allele a.

MC 60. What type of growth equation is $dN/dt = rN(K-N)/K$?

A. Exponential growth
B. Linear growth
C. Logistic growth
D. Geometric growth

Correct answer: C

Rationale: This is the logistic growth equation, which includes carrying capacity (K) as a limiting factor.

MC 61. According to the material, what creates selective pressures?

A. Stable allele frequencies
B. Environmental conditions and species interactions
C. Hardy-Weinberg equilibrium
D. Absence of competition

Correct answer: B

Rationale: Environmental conditions and species interactions create selective pressures that drive evolution.

MC 62. What happens to heterozygote frequency if mating is nonrandom but allele frequencies stay constant?

A. It remains at 2pq

B. It can deviate from 2pq

C. It always increases

D. It becomes zero

Correct answer: B

Rationale: Nonrandom mating can alter genotype frequencies (including heterozygotes) even if allele frequencies remain stable.

MC 63. What mathematical tool helps understand population patterns according to the material?

A. Only observation

B. Mathematical models like Hardy-Weinberg equations

C. Guesswork

D. Random sampling

Correct answer: B

Rationale: Mathematical models including Hardy-Weinberg equations and population growth formulas provide quantitative tools for understanding biological patterns.

MC 64. In resource partitioning, what allows species coexistence?

A. Identical resource use

B. Complete competition

C. Different resource use reducing direct competition

D. Absence of resources

Correct answer: C

Rationale: Resource partitioning allows species coexistence through different resource use that reduces direct competition.

MC 65. What indicates that evolutionary forces are acting on a population?

A. Exact Hardy-Weinberg predictions observed

B. Deviation from Hardy-Weinberg predictions

C. Stable population size

D. Equal allele frequencies

Correct answer: B

Rationale: Any significant deviation from Hardy-Weinberg predictions indicates that one or more evolutionary forces is acting on the population.

MC 66. According to the material, what operates simultaneously in populations?

A. Only evolutionary processes

B. Only ecological processes

C. Evolutionary and ecological processes

D. Neither process

Correct answer: C

Rationale: Evolutionary and ecological processes operate simultaneously and often influence each other.

MC 67. What happens when predator populations are high according to the predator-prey dynamics?

A. Prey populations increase

B. Prey populations are suppressed

C. No effect on prey

D. Both populations crash

Correct answer: B

Rationale: When predator populations are high, prey populations are suppressed through increased predation.

MC 68. What represents the maximum sustainable population size?

A. Growth rate (r)
B. Birth rate
C. Carrying capacity (K)
D. Death rate

Correct answer: C

Rationale: Carrying capacity (K) represents the maximum sustainable population size in an environment.

MC 69. According to Hardy-Weinberg, what is the probability of forming an Aa genotype?

A. $p \times q$
B. $2pq$
C. $p^2 + q^2$
D. pq

Correct answer: B

Rationale: The probability of forming Aa equals $(p \times q) + (q \times p) = 2pq$ due to two ways of forming heterozygotes.

MC 70. What time scales must be integrated to understand population dynamics according to the material?

A. Only short-term responses
B. Only long-term evolution
C. Both immediate demographic responses and long-term evolutionary change
D. Neither time scale

Correct answer: C

Rationale: Understanding population and community dynamics requires integrating multiple time scales, from immediate demographic responses to long-term evolutionary change.

MC 71. What connects individual-level traits to ecosystem patterns?

A. Random chance only
B. Species interactions and environmental pressures
C. Isolation of traits
D. Absence of evolution

Correct answer: B

Rationale: Species interactions and environmental pressures shape both individual traits and ecosystem-level patterns.

MC 72. In a population with 160 dominant homozygotes, 480 heterozygotes, and 360 recessive homozygotes, is it in Hardy-Weinberg equilibrium if $p = 0.4$?

A. Yes, these are the exact predicted numbers
B. No, there are too many heterozygotes
C. No, there are too few recessive homozygotes
D. Cannot determine from this information

Correct answer: A

Rationale: With $p = 0.4$ and $q = 0.6$ in a population of 1000: $p^2 = 0.16$ (160), $2pq = 0.48$ (480), $q^2 = 0.36$ (360). These match exactly.

MC 73. What creates the oscillating pattern in predator-prey populations?

A. Random environmental changes
B. Time lag between prey abundance and predator response
C. Constant population sizes
D. Absence of interaction

Correct answer: B

Rationale: The time lag between changes in prey abundance and predator population response creates characteristic oscillations.

MC 74. According to the material, what shapes ecosystem structure over multiple time scales?

A. Only abiotic factors
B. Feedback loops between population, community, and ecosystem levels
C. Single species effects
D. Isolation of each level

Correct answer: B

Rationale: Feedback loops between population, community, and ecosystem levels shape ecosystem structure and function over multiple time scales.

MC 75. What must researchers maintain to prevent Hardy-Weinberg violations in laboratory Drosophila populations?

A. Small population sizes
B. High mutation rates
C. Large population sizes and controlled conditions
D. Selective breeding

Correct answer: C

Rationale: Researchers must maintain large population sizes and control other conditions to prevent violations of Hardy-Weinberg equilibrium.

Unit D Lesson 1 Review

Constructed Response Bank

Question 1

A student claims that 'Hardy-Weinberg equilibrium means all populations have equal frequencies of dominant and recessive alleles.'

a. State whether this claim is correct or incorrect and explain what Hardy-Weinberg equilibrium actually represents.

b. Explain how the mathematical relationship $p^2 + 2pq + q^2 = 1$ emerges from random mating.

c. Identify one condition that must be met for Hardy-Weinberg equilibrium to occur.

Question 2

Researchers observe that a Drosophila population has 360 white-eyed flies out of 1000 total, where white eyes are recessive.

a. Calculate the frequency of both alleles in this population.

b. Predict the number of heterozygous flies expected if the population is in Hardy-Weinberg equilibrium.

c. Name one evolutionary force that would cause deviation from these predicted numbers.

Question 3

A marine biologist observes that fish populations show logistic growth patterns after establishing a protected area.

a. Describe the three phases of logistic growth and what characterizes each phase.

b. Explain how the term $(K-N)/K$ in the logistic equation creates the S-shaped curve.

c. State one factor that determines carrying capacity in marine ecosystems.

Question 4

A researcher states that 'Gene flow between populations always disrupts Hardy-Weinberg equilibrium.'

a. Evaluate whether this statement is accurate and explain why.

b. Describe how gene flow differs from mutation in its effect on population genetics.

c. Identify one way gene flow could increase fitness in a small population.

Question 5

Ecologists observe predator-prey cycles between lynx and snowshoe hares showing regular oscillations.

a. Explain the mechanism that creates oscillating population patterns in predator-prey relationships.

b. Describe how these oscillations represent negative feedback loops.

c. State one factor besides predation that could affect hare population cycles.

Question 6

A student observes that r-selected species dominate after forest fires while K-selected species dominate in old-growth forests.

a. Compare the reproductive strategies of r-selected and K-selected species.

b. Explain why r-selected species are favored in disturbed habitats like post-fire environments.

c. Provide one example of an r-selected species characteristic.

Question 7

Research shows that nonrandom mating changes genotype frequencies but can leave allele frequencies unchanged.

a. Explain how nonrandom mating affects genotype frequencies differently than allele frequencies.

b. Describe why this pattern violates Hardy-Weinberg equilibrium despite stable allele frequencies.

c. Name one type of nonrandom mating that occurs in natural populations.

Question 8

Scientists discover that small populations experience more dramatic allele frequency changes than large populations.

a. Explain the mechanism of genetic drift and why it affects small populations more strongly.

b. Describe how genetic drift differs from natural selection in changing allele frequencies.

c. State one consequence of genetic drift for conservation of endangered species.

Question 9

Observations show that competitive exclusion leads to resource partitioning in similar species.

a. State the competitive exclusion principle and explain its biological significance.

b. Explain how resource partitioning allows species coexistence despite competition.

c. Provide one example of how species might partition resources.

Question 10

A population shows exponential growth initially but then transitions to logistic growth as it approaches carrying capacity.

a. Compare the mathematical equations for exponential and logistic growth, explaining each term.

b. Explain what environmental factors create the transition from exponential to logistic growth.

c. Name one assumption of exponential growth that becomes violated as populations grow.

Question 11

Researchers find that keystone species have disproportionate effects on community structure relative to their abundance.

a. Define keystone species and explain why their impact exceeds their abundance.

b. Describe the community-level consequences of removing a keystone species.

c. Identify one mechanism by which a predator can act as a keystone species.

Question 12

Field studies reveal that primary succession takes much longer than secondary succession after disturbance.

a. Distinguish between primary and secondary succession, explaining the key difference.

b. Explain why secondary succession proceeds more rapidly than primary succession.

c. Name one type of disturbance that leads to secondary succession.

Question 13

A biology student claims that 'Mutations alone can maintain Hardy-Weinberg equilibrium because they create genetic variation.'

a. Evaluate this claim and explain the actual role of mutations in Hardy-Weinberg equilibrium.

b. Explain how mutations differ from other evolutionary forces in their rate of allele frequency change.

c. State one reason why mutations are important for evolution despite their low rate.

Question 14

Ecologists observe that mutualistic relationships between species create positive feedback loops affecting community structure.

a. Define mutualistic relationships and explain how both species benefit.

b. Explain how mutualistic relationships can create positive feedback loops in communities.

c. Provide one example of a mutualistic relationship from the material or general knowledge.

Question 15

Laboratory studies show that starting Drosophila populations with $p = q = 0.5$ results in predictable genotype frequencies if Hardy-Weinberg conditions are met.

a. Calculate the expected genotype frequencies when $p = q = 0.5$.

b. Explain why laboratory populations are useful for testing Hardy-Weinberg predictions.

c. Name one challenge in maintaining Hardy-Weinberg conditions in laboratory populations.

Question 16

Research indicates that Type III survivorship curves characterize species with high reproductive output but minimal parental care.

a. Describe the shape and characteristics of Type III survivorship curves.

b. Explain the relationship between Type III curves and r-selected reproductive strategies.

c. Name one organism that typically shows a Type III survivorship curve.

Question 17

A protected coral reef shows increased genetic diversity in fish populations after decades of heavy fishing pressure was removed.

a. Explain how heavy fishing acts as a selective pressure affecting genetic diversity.

b. Describe the mechanisms by which protection increases genetic diversity over time.

c. State one ecological benefit of increased genetic diversity in fish populations.

Question 18

Scientists observe that environmental conditions and species interactions create cascading effects across biological organization levels.

a. Explain what cascading effects are and how they connect different organizational levels.

b. Describe how selective pressures link population genetics to community ecology.

c. Provide one example of a cascading effect from genes to ecosystems.

Question 19

Population biologists note that understanding dynamics requires integrating immediate demographic responses with long-term evolutionary changes.

a. Distinguish between demographic and evolutionary responses to environmental change.

b. Explain why both time scales must be considered for population predictions.

c. Name one factor that operates on both demographic and evolutionary time scales.

Question 20

Field observations show that life history strategies represent coordinated trait combinations optimized for specific environments.

a. Define life history strategy and explain why traits are coordinated rather than independent.

b. Explain how environmental stability influences the evolution of life history strategies.

c. Identify one trade-off that constrains life history evolution.

Model Answers & Rubrics

Question 1

Part a: State whether this claim is correct or incorrect and explain what Hardy-Weinberg equilibrium actually represents.

Model Answer: The claim is incorrect. Hardy-Weinberg equilibrium does not require equal allele frequencies ($p = q = 0.5$). It represents a state where allele frequencies and genotype frequencies remain constant across generations when no evolutionary forces are acting. Any allele frequencies can be in equilibrium as long as the five conditions are met. (2 marks) (2 marks)

Part b: Explain how the mathematical relationship $p^2 + 2pq + q^2 = 1$ emerges from random mating.

Model Answer: During random mating, gametes combine randomly based on allele frequencies in the gene pool. The probability of forming AA = $p \times p = p^2$, aa = $q \times q = q^2$, and Aa = $(p \times q) + (q \times p) = 2pq$. These probabilities emerge automatically through random combination of gametes during fertilization, with no special mechanisms required. (2 marks) (2 marks)

Part c: Identify one condition that must be met for Hardy-Weinberg equilibrium to occur.

Model Answer: Large population size (to prevent genetic drift). [Also acceptable: no mutations, no gene flow, random mating, or no natural selection] (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for correctly identifying claim as incorrect, 1 mark for accurate explanation of equilibrium

Part b (2 marks): 1 mark for explaining random gamete combination, 1 mark for showing how this creates the equation

Part c (1 mark): 1 mark for correctly identifying any one of the five conditions

Question 2

Part a: Calculate the frequency of both alleles in this population.

Model Answer: Since 360/1000 flies are white-eyed (recessive), $q^2 = 0.36$, so $q = 0.6$. Using $p + q = 1$, $p = 1 - 0.6 = 0.4$. The dominant allele frequency is 0.4 and recessive is 0.6. (2 marks) (2 marks)

Part b: Predict the number of heterozygous flies expected if the population is in Hardy-Weinberg equilibrium.

Model Answer: Heterozygote frequency = $2pq = 2(0.4)(0.6) = 0.48$. In a population of 1000, this equals $0.48 \times 1000 = 480$ heterozygous flies. (2 marks) (2 marks)

Part c: Name one evolutionary force that would cause deviation from these predicted numbers.

Model Answer: Natural selection [Also acceptable: genetic drift, gene flow, mutations, or nonrandom mating] (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for correct calculation of q, 1 mark for correct calculation of p

Part b (2 marks): 1 mark for correct 2pq calculation, 1 mark for converting to number of individuals

Part c (1 mark): 1 mark for naming any valid evolutionary force

Question 3

Part a: Describe the three phases of logistic growth and what characterizes each phase.

Model Answer: Lag phase: slow initial growth as individuals establish and begin reproducing. Exponential phase: rapid population growth when resources are abundant. Plateau phase: growth slows and stops as population reaches carrying capacity (K) due to environmental resistance. (2 marks) (2 marks)

Part b: Explain how the term (K-N)/K in the logistic equation creates the S-shaped curve.

Model Answer: When N is small, (K-N)/K is close to 1, allowing near-exponential growth. As N approaches K, (K-N)/K approaches 0, reducing growth rate. This creates deceleration that forms the S-shape as environmental resistance increases with population size. (2 marks) (2 marks)

Part c: State one factor that determines carrying capacity in marine ecosystems.

Model Answer: Food availability [Also acceptable: space/territory, dissolved oxygen, nesting sites, or predation pressure] (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for naming three phases, 1 mark for characterizing each correctly

Part b (2 marks): 1 mark for explaining effect when N is small, 1 mark for explaining effect as N approaches K

Part c (1 mark): 1 mark for valid limiting factor

Question 4

Part a: Evaluate whether this statement is accurate and explain why.

Model Answer: The statement is accurate. Gene flow brings in alleles from other populations, changing allele frequencies in the recipient population. This violates one of the five required conditions for Hardy-Weinberg equilibrium (no gene flow), causing deviation from equilibrium predictions. (2 marks) (2 marks)

Part b: Describe how gene flow differs from mutation in its effect on population genetics.

Model Answer: Gene flow introduces existing alleles from other populations, potentially causing large immediate changes in allele frequencies. Mutations create new alleles or alter existing ones at very low rates, causing slower changes. Gene flow affects multiple alleles simultaneously while mutations typically affect single alleles. (2 marks) (2 marks)

Part c: Identify one way gene flow could increase fitness in a small population.

Model Answer: Introduces genetic variation that may have been lost to genetic drift. [Also acceptable: reduces inbreeding, introduces beneficial alleles] (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for confirming accuracy, 1 mark for explaining mechanism

Part b (2 marks): 1 mark for describing gene flow effects, 1 mark for contrasting with mutation

Part c (1 mark): 1 mark for valid benefit

Question 5

Part a: Explain the mechanism that creates oscillating population patterns in predator-prey relationships.

Model Answer: When prey are abundant, predator populations increase due to available food. High predator numbers suppress prey populations through increased predation. As prey decline, predators crash from lack of food. Low predator numbers allow prey recovery, restarting the cycle. The time lag between prey changes and predator response creates oscillations. (2 marks) (2 marks)

Part b: Describe how these oscillations represent negative feedback loops.

Model Answer: Negative feedback occurs when an increase in one population (prey) leads to changes (predator increase) that eventually reverse the initial increase (prey decrease). Similarly, predator increases lead to their own eventual decrease through prey depletion. This self-regulating mechanism prevents either population from growing indefinitely. (2 marks) (2 marks)

Part c: State one factor besides predation that could affect hare population cycles.

Model Answer: Food availability/plant cycles [Also acceptable: disease, weather/climate, territorial behavior, stress] (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for describing sequential changes, 1 mark for explaining time lag effect

Part b (2 marks): 1 mark for defining negative feedback, 1 mark for applying to predator-prey

Part c (1 mark): 1 mark for valid additional factor

Question 6

Part a: Compare the reproductive strategies of r-selected and K-selected species.

Model Answer: r-selected species produce many offspring with minimal parental investment, mature quickly, and have short lifespans. K-selected species produce few offspring with substantial parental investment, mature slowly, and have longer lifespans. r-selected maximize reproductive rate while K-selected maximize competitive ability. (2 marks) (2 marks)

Part b: Explain why r-selected species are favored in disturbed habitats like post-fire environments.

Model Answer: Disturbed habitats have abundant resources and little competition initially. r-selected species' rapid reproduction and early maturity allow them to quickly colonize and exploit these resources before competition intensifies. Their high reproductive output compensates for high mortality in unstable conditions. (2 marks) (2 marks)

Part c: Provide one example of an r-selected species characteristic.

Model Answer: Large number of seeds/offspring [Also acceptable: small body size, early maturity, short generation time, high growth rate] (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for r-selected traits, 1 mark for K-selected traits

Part b (2 marks): 1 mark for describing disturbed habitat conditions, 1 mark for explaining r-selected advantage

Part c (1 mark): 1 mark for valid characteristic

Question 7

Part a: Explain how nonrandom mating affects genotype frequencies differently than allele frequencies.

Model Answer: Nonrandom mating changes the way alleles combine into genotypes without changing the total number of each allele in the population. For example, inbreeding increases homozygote frequency and decreases heterozygote frequency, but the overall proportion of each allele remains the same. (2 marks) (2 marks)

Part b: Describe why this pattern violates Hardy-Weinberg equilibrium despite stable allele frequencies.

Model Answer: Hardy-Weinberg equilibrium requires both allele AND genotype frequencies to remain constant. Even if p and q don't change, deviation from the expected $p^2 + 2pq + q^2 = 1$ distribution means the population is not in equilibrium. Random mating is one of the five required conditions. (2 marks) (2 marks)

Part c: Name one type of nonrandom mating that occurs in natural populations.

Model Answer: Inbreeding [Also acceptable: assortative mating, disassortative mating, sexual selection] (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for explaining genotype changes, 1 mark for explaining allele stability

Part b (2 marks): 1 mark for identifying both frequencies must be stable, 1 mark for connecting to equilibrium conditions

Part c (1 mark): 1 mark for valid type

Question 8

Part a: Explain the mechanism of genetic drift and why it affects small populations more strongly.

Model Answer: Genetic drift is random sampling error in allele transmission between generations. In small populations, random events (deaths, mating success) have proportionally larger effects on allele frequencies. Large populations buffer against these random effects through statistical averaging. (2 marks) (2 marks)

Part b: Describe how genetic drift differs from natural selection in changing allele frequencies.

Model Answer: Genetic drift causes random changes regardless of fitness effects, potentially fixing harmful alleles or losing beneficial ones. Natural selection causes directional changes based on fitness advantages, consistently favoring beneficial alleles. Drift is stronger in small populations while selection can be strong regardless of size. (2 marks) (2 marks)

Part c: State one consequence of genetic drift for conservation of endangered species.

Model Answer: Loss of genetic diversity [Also acceptable: increased inbreeding, reduced adaptive potential, fixation of deleterious alleles] (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for defining drift, 1 mark for explaining size effect

Part b (2 marks): 1 mark for describing drift as random, 1 mark for describing selection as fitness-based

Part c (1 mark): 1 mark for valid conservation consequence

Question 9

Part a: State the competitive exclusion principle and explain its biological significance.

Model Answer: The competitive exclusion principle states that two species with identical niches cannot coexist; one will eliminate the other through superior competitive ability. This drives niche differentiation and species diversification, shaping community structure and promoting biodiversity through resource partitioning. (2 marks) (2 marks)

Part b: Explain how resource partitioning allows species coexistence despite competition.

Model Answer: Resource partitioning reduces direct competition by dividing resources along different dimensions (time, space, size). Species evolve to use different portions of available resources, reducing niche overlap. This allows multiple species to coexist by minimizing competitive interactions. (2 marks) (2 marks)

Part c: Provide one example of how species might partition resources.

Model Answer: Different feeding times (diurnal vs nocturnal) [Also acceptable: different food sizes, different habitat zones, different seasons] (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for stating principle, 1 mark for explaining significance

Part b (2 marks): 1 mark for describing partitioning mechanism, 1 mark for explaining coexistence

Part c (1 mark): 1 mark for valid example

Question 10

Part a: Compare the mathematical equations for exponential and logistic growth, explaining each term.

Model Answer: Exponential: $dN/dt = rN$, where r is per capita growth rate and N is population size. Growth is proportional to population size only. Logistic: $dN/dt = rN(K-N)/K$, which adds the term $(K-N)/K$ where K is carrying capacity. This term reduces growth as N approaches K. (2 marks) (2 marks)

Part b: Explain what environmental factors create the transition from exponential to logistic growth.

Model Answer: Initially unlimited resources allow exponential growth. As population increases, intraspecific competition for limited resources (food, space, mates) intensifies. This environmental resistance increases with density, reducing birth rates and/or increasing death rates, causing the transition to logistic growth. (2 marks) (2 marks)

Part c: Name one assumption of exponential growth that becomes violated as populations grow.

Model Answer: Unlimited resources [Also acceptable: no competition, constant birth/death rates, unlimited space] (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for each equation with term explanations

Part b (2 marks): 1 mark for describing resource limitation, 1 mark for explaining density-dependent effects

Part c (1 mark): 1 mark for valid assumption

Question 11

Part a: Define keystone species and explain why their impact exceeds their abundance.

Model Answer: Keystone species are organisms whose effects on community structure are much larger than expected from their population size. They often control populations of dominant competitors, maintain diversity, or provide critical resources, creating cascading effects throughout the community food web. (2 marks) (2 marks)

Part b: Describe the community-level consequences of removing a keystone species.

Model Answer: Removal causes dramatic shifts in species composition and abundance. Previously controlled species may explode in number, competitively excluding others. This reduces diversity, alters trophic structure, and can trigger cascading effects that fundamentally change ecosystem functioning. (2 marks) (2 marks)

Part c: Identify one mechanism by which a predator can act as a keystone species.

Model Answer: Preventing competitive exclusion by preying on dominant competitors [Also acceptable: maintaining prey diversity, controlling herbivore populations] (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for definition, 1 mark for explaining disproportionate impact

Part b (2 marks): 1 mark for describing immediate changes, 1 mark for cascading effects

Part c (1 mark): 1 mark for valid mechanism

Question 12

Part a: Distinguish between primary and secondary succession, explaining the key difference.

Model Answer: Primary succession begins on bare substrate with no soil (volcanic rock, glacial till), requiring pioneer species to build soil from scratch. Secondary succession occurs after disturbance where soil remains intact, allowing faster recolonization. The presence or absence of soil is the key difference. (2 marks) (2 marks)

Part b: Explain why secondary succession proceeds more rapidly than primary succession.

Model Answer: Secondary succession has existing soil with nutrients, organic matter, and seed bank. Soil contains dormant seeds and root systems that can resprout. The established soil structure supports immediate plant growth, while primary succession must first weather rock and accumulate organic matter over centuries. (2 marks) (2 marks)

Part c: Name one type of disturbance that leads to secondary succession.

Model Answer: Forest fire [Also acceptable: hurricane, logging, abandonment of agricultural land, flooding] (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for defining each type, 1 mark for identifying soil as key difference

Part b (2 marks): 1 mark for soil advantages, 1 mark for contrasting with primary succession timing

Part c (1 mark): 1 mark for valid disturbance

Question 13

Part a: Evaluate this claim and explain the actual role of mutations in Hardy-Weinberg equilibrium.

Model Answer: The claim is incorrect. Mutations violate Hardy-Weinberg equilibrium by introducing new alleles or altering existing ones, changing allele frequencies. One of the five required conditions for equilibrium is 'no mutations.' While mutations create variation, they prevent equilibrium from being maintained. (2 marks) (2 marks)

Part b: Explain how mutations differ from other evolutionary forces in their rate of allele frequency change.

Model Answer: Mutations occur at very low rates (typically 10^-8 to 10^-9 per generation), causing extremely slow allele frequency changes. Other forces like selection, drift, or gene flow can cause rapid changes within a few generations. Mutations are the ultimate source of variation but weak as a direct evolutionary force. (2 marks) (2 marks)

Part c: State one reason why mutations are important for evolution despite their low rate.

Model Answer: They are the only source of new alleles [Also acceptable: provide raw material for selection, maintain genetic variation] (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for identifying incorrect claim, 1 mark for explaining violation of equilibrium

Part b (2 marks): 1 mark for mutation rate, 1 mark for comparing to other forces

Part c (1 mark): 1 mark for valid importance

Question 14

Part a: Define mutualistic relationships and explain how both species benefit.

Model Answer: Mutualistic relationships are interactions where both species benefit from the association. Benefits can include resource exchange (nutrients for carbohydrates), protection services, dispersal assistance, or improved reproduction. Each species provides something the other needs while receiving reciprocal benefits. (2 marks) (2 marks)

Part b: Explain how mutualistic relationships can create positive feedback loops in communities.

Model Answer: As one mutualistic partner increases in abundance, it provides more benefits to its partner, allowing that species to also increase. This mutual enhancement creates positive feedback where both populations grow together, potentially altering community composition and resource availability for other species. (2 marks) (2 marks)

Part c: Provide one example of a mutualistic relationship from the material or general knowledge.

Model Answer: Plant-pollinator relationships [Also acceptable: mycorrhizae-plant, nitrogen-fixing bacteria-legumes, cleaner fish-host fish] (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for definition, 1 mark for explaining mutual benefits

Part b (2 marks): 1 mark for describing enhancement mechanism, 1 mark for community effects

Part c (1 mark): 1 mark for valid example

Question 15

Part a: Calculate the expected genotype frequencies when $p = q = 0.5$.

Model Answer: When $p = q = 0.5$: p^2 (homozygous dominant) = $(0.5)^2 = 0.25$, $2pq$ (heterozygous) = $2(0.5)(0.5) = 0.50$, q^2 (homozygous recessive) = $(0.5)^2 = 0.25$. Expected ratio is 1:2:1 or 25%:50%:25%. (2 marks) (2 marks)

Part b: Explain why laboratory populations are useful for testing Hardy-Weinberg predictions.

Model Answer: Laboratory conditions allow control of the five equilibrium conditions: preventing mutations through short timeframes, eliminating gene flow through isolation, maintaining large population sizes, ensuring random mating, and removing selection pressures. This control allows testing of mathematical predictions. (2 marks) (2 marks)

Part c: Name one challenge in maintaining Hardy-Weinberg conditions in laboratory populations.

Model Answer: Preventing genetic drift in finite populations [Also acceptable: eliminating all selection, ensuring truly random mating, preventing mutations] (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for correct calculations, 1 mark for all three frequencies

Part b (2 marks): 1 mark for control advantage, 1 mark for connecting to conditions

Part c (1 mark): 1 mark for valid challenge

Question 16

Part a: Describe the shape and characteristics of Type III survivorship curves.

Model Answer: Type III curves show extremely high mortality early in life with few individuals surviving to adulthood. The curve drops steeply initially then levels off for survivors. This pattern reflects species producing many offspring with high juvenile mortality but relatively stable adult survival. (2 marks) (2 marks)

Part b: Explain the relationship between Type III curves and r-selected reproductive strategies.

Model Answer: r-selected species produce many offspring with minimal parental investment, accepting high juvenile mortality. This strategy matches Type III curves where most offspring die young. The high reproductive output compensates for low survival probability, maintaining population despite massive early losses. (2 marks) (2 marks)

Part c: Name one organism that typically shows a Type III survivorship curve.

Model Answer: Fish [Also acceptable: frogs, marine invertebrates, many plants, oysters, sea turtles] (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for describing shape, 1 mark for mortality pattern

Part b (2 marks): 1 mark for connecting to r-selection, 1 mark for explaining compensation strategy

Part c (1 mark): 1 mark for valid organism

Question 17

Part a: Explain how heavy fishing acts as a selective pressure affecting genetic diversity.

Model Answer: Heavy fishing selectively removes certain phenotypes (larger individuals, bold behavior), reducing alleles associated with these traits. This creates directional selection that reduces genetic variation. Overfishing can also reduce population size, increasing genetic drift effects that further erode diversity. (2 marks) (2 marks)

Part b: Describe the mechanisms by which protection increases genetic diversity over time.

Model Answer: Protection removes selective pressure, allowing previously disadvantageous alleles to persist. Larger population sizes reduce genetic drift. Increased survival allows more genetic variants to reproduce. Immigration from other areas introduces new alleles through gene flow, restoring lost variation. (2 marks) (2 marks)

Part c: State one ecological benefit of increased genetic diversity in fish populations.

Model Answer: Greater adaptability to environmental change [Also acceptable: disease resistance, ecosystem stability, evolutionary potential] (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for selection effects, 1 mark for population size effects

Part b (2 marks): 1 mark for removing selection, 1 mark for gene flow/drift reduction

Part c (1 mark): 1 mark for valid benefit

Question 18

Part a: Explain what cascading effects are and how they connect different organizational levels.

Model Answer: Cascading effects are chains of cause and effect where changes at one biological level trigger changes at other levels. For example, environmental change affects gene frequencies, which alters population dynamics, changing species interactions, modifying community structure, and ultimately affecting ecosystem function. (2 marks) (2 marks)

Part b: Describe how selective pressures link population genetics to community ecology.

Model Answer: Environmental conditions and species interactions create selective pressures that change allele frequencies through natural selection. These genetic changes alter traits affecting competitive ability, predator avoidance, or resource use. Modified traits change species interaction outcomes, restructuring community relationships and species abundances. (2 marks) (2 marks)

Part c: Provide one example of a cascading effect from genes to ecosystems.

Model Answer: Evolution of pesticide resistance altering agricultural ecosystem dynamics [Also acceptable: predator adaptation affecting prey communities, evolution of disease resistance] (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for defining cascading effects, 1 mark for showing connections

Part b (2 marks): 1 mark for genetic changes, 1 mark for community consequences

Part c (1 mark): 1 mark for valid example

Question 19

Part a: Distinguish between demographic and evolutionary responses to environmental change.

Model Answer: Demographic responses are immediate changes in birth rates, death rates, and population size without genetic change. Evolutionary responses involve changes in allele frequencies over generations through selection, creating heritable trait changes. Demographics occur within generations while evolution occurs across generations. (2 marks) (2 marks)

Part b: Explain why both time scales must be considered for population predictions.

Model Answer: Short-term demographic changes determine immediate population survival and trajectory. Long-term evolutionary changes determine adaptation potential and future demographic parameters. Ignoring either scale misses critical feedback: evolution affects demographic rates while population size influences evolutionary processes like drift. (2 marks) (2 marks)

Part c: Name one factor that operates on both demographic and evolutionary time scales.

Model Answer: Predation pressure [Also acceptable: resource availability, climate, disease, competition] (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for demographic definition, 1 mark for evolutionary definition

Part b (2 marks): 1 mark for importance of each scale, 1 mark for interaction between scales

Part c (1 mark): 1 mark for valid factor

Question 20

Part a: Define life history strategy and explain why traits are coordinated rather than independent.

Model Answer: Life history strategy is the pattern of survival and reproduction traits evolved to maximize fitness in particular environments. Traits are coordinated because energy/resource allocation creates trade-offs: investment in one trait reduces resources for others. Natural selection favors trait combinations that work together efficiently. (2 marks) (2 marks)

Part b: Explain how environmental stability influences the evolution of life history strategies.

Model Answer: Stable environments favor K-selected strategies: slower development, fewer offspring with greater investment, competing for limited resources near carrying capacity. Unstable environments favor r-selected strategies: rapid reproduction, many offspring, exploiting temporary resource abundance. Environmental predictability determines optimal resource allocation patterns. (2 marks) (2 marks)

Part c: Identify one trade-off that constrains life history evolution.

Model Answer: Number versus size of offspring [Also acceptable: current versus future reproduction, growth versus reproduction, survival versus reproduction] (1 mark) (1 marks)

Scoring rubric

- Part a (2 marks): 1 mark for definition, 1 mark for explaining coordination
- Part b (2 marks): 1 mark for stable environment strategy, 1 mark for unstable environment strategy
- Part c (1 mark): 1 mark for valid trade-off